Strategic Fee Choice and Transaction-Priority Allocation under Incomplete Information

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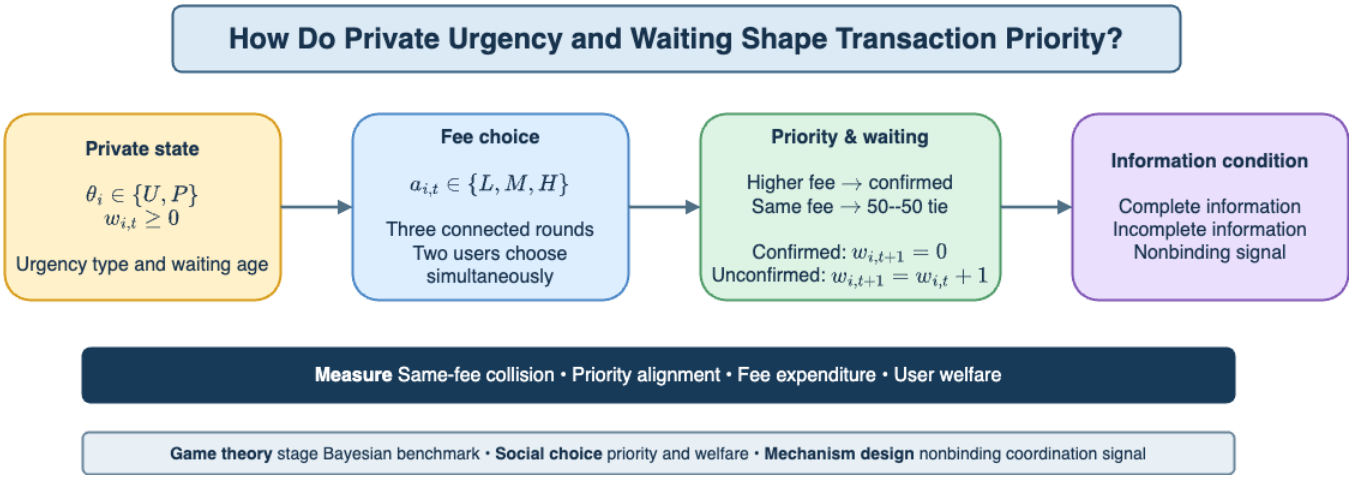


Figure 1: Research design. Two users with private Urgent or Patient types choose Low, Medium, or High fees across three connected rounds. Higher fees receive confirmation priority, while equal fees use a random tie-break and unconfirmed users accumulate waiting. We compare complete information, incomplete information, and incomplete information with a nonbinding coordination signal, evaluating priority allocation, fee expenditure, and user welfare.

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1 Three connected research questions and verified gap

Blockchain systems must allocate limited transaction-processing capacity among users with different incentives, creating design questions for users, wallet designers, and protocol developers [6]. We study a simplified fee-choice setting motivated by Algorand [3]. Two users interact for three rounds and choose a Low (L), Medium (M), or High (H) fee. Each privately knows whether the transaction is Urgent or Patient. A higher fee receives confirmation priority; equal fees use a random tie-break. An unconfirmed user accumulates waiting, which changes incentives in later rounds.

Our shared question is how private urgency and accumulated waiting affect transaction-priority allocation, and whether a coordination signal can improve it. **Q1—Economics:** How do urgency, waiting, and fee costs shape strategic choices and welfare? **Q2—Computation:** How do information and coordination rules change allocation outcomes? **Q3—Behavior:** How do users' beliefs about the opponent affect fee choices and responses to recommendations?

Bayesian game theory provides a framework for strategic decisions under private information [5], while transaction-fee mechanism design studies allocation of scarce blockchain capacity [6]. Our project connects these questions in a dynamic setting where private types, discrete fee choices, and accumulated waiting jointly determine priority. Neither strategic prediction, computational allocation, nor behavioral choice alone resolves how the mechanism performs.

2 Answer to Q1: incentives and strategic model

There are two users $i \in \{1, 2\}$ with private types $\theta_i \in \{U, P\}$, where U is Urgent and P is Patient. In each round $t \in \{1, 2, 3\}$, both simultaneously choose

$$a_{i,t} \in \{L, M, H\},$$

with $c(L) = 1$, $c(M) = 2$, and $c(H) = 3$. We use $(V_U, d_U) = (8, 2)$ and $(V_P, d_P) = (5, 1)$, where V_θ is confirmation value and d_θ is delay cost.

The higher fee receives confirmation priority; equal fees use a 50–50 tie-break. If $w_{i,t}$ is waiting age before round t,

$$w_{i,t+1} = \begin{cases} 0, & \text{if confirmed,} \\ w_{i,t} + 1, & \text{otherwise,} \end{cases} \qquad u_{i,t} = \begin{cases} V_{\theta_i} - c(a_{i,t}), & \text{if confirmed,} \\ -d_{\theta_i} w_{i,t+1}, & \text{otherwise.} \end{cases}$$

The full setting is a finite-horizon dynamic Bayesian game. For a tractable game-theoretic prediction, we use a Bayesian Nash best-response benchmark at each simultaneous fee-choice stage [5]. Against H, choosing L or M gives $-d_\theta(w+1)$, while

$$EU_\theta(H \mid H, w) = \frac{1}{2}[V_\theta - c(H)] - \frac{1}{2}d_\theta(w+1).$$

Thus H is a strict best response to H when

$$V_\theta - c(H) + d_\theta(w+1) > 0.$$

This holds for both types under our calibration, so (H, H) is a stage Bayesian Nash equilibrium. The prediction identifies the strategic tension: individually rational fee escalation can produce a same-fee collision instead of differentiated priority.

Game theory provides this strategic benchmark. Social choice evaluates whether scarce priority is aligned with urgency and the trade-off between allocation quality and user welfare. Mechanism design introduces differentiated, nonbinding recommendations and tests whether they improve allocation while choices remain voluntary.

3 Answer to Q2: computation, allocation comparison, and artifacts

The Colab implements the same three-round game. Its state-responsive comparison policy raises fees with urgency or waiting:

$$\pi(U, w) = \begin{cases} M, & w = 0, \\ H, & w \geq 1, \end{cases} \qquad \pi(P, w) = \begin{cases} L, & w = 0, \\ M, & w = 1, \\ H, & w \geq 2. \end{cases}$$

This is a simulation policy rather than the equilibrium prediction above.

We compare the same priority-allocation environment under three information and coordination conditions. The complete-information benchmark coordinates differentiated recommendations using both users' types and waiting states. Under incomplete information, each user follows the comparison policy using only its own state. With a signal, private types remain hidden from the other user while the system uses both users' types and waiting states to provide differentiated, nonbinding recommendations based on

$$q_{i,t} = 2\mathbf{1}\{\theta_i = U\} + w_{i,t},$$

with baseline compliance $\rho = 0.85$. The simulation samples types, applies the relevant policy or recommendation, allocates confirmation to the higher fee with random tie-breaking, and updates waiting for the next round.

The scarce resource is transaction confirmation priority. Fee choice determines priority, and the confirmed user's fee expenditure is the payment measure. We report same-fee collision, priority alignment, urgent-user confirmation, waiting, fee expenditure, individual utility, and total user welfare. In 100,000 simulated sessions, the signal reduces same-fee collision from 33.36% to 5.20% and raises priority alignment from 87.11% to 97.99%. Urgent-user confirmation rises from 66.39% to 81.25%, but mean fee expenditure rises from 2.352 to 2.812 and mean total user welfare per round falls from 3.057 to 2.678. Thus the signal improves priority allocation at the cost of higher fee expenditure, rather than producing an automatic welfare improvement. Signal-compliance sensitivity confirms the same trade-off.

Computational scientist—GitHub:
https://github.com/dku-comsci-econ206-Autumn2026/PS2-FP7-ZijieWang-LinZhang
Runnable notebook:
https://colab.research.google.com/drive/1E5NDNUU62rHPwp9UCG9oEHXplz3PG9EJ?usp=sharing
Behavioral scientist—Hugging Face:
https://huggingface.co/spaces/dku-comsci-econ206-2026/SubmitOrWait

4 Answer to Q3: behavior and interdisciplinary integration

The rational benchmark predicts choosing H when a user expects the opponent to choose H. Actual choices may differ because of strategic uncertainty, bounded reasoning, or learning from previous outcomes [1]. The Hugging Face game implements the same three-round $L/M/H$ setting and records each player's prediction of the opponent's action, confidence, reasoning, and fee choice.

This design provides a discriminating test. Choices moving toward H when players expect H support the best-response benchmark; systematic responses to accumulated waiting support state-dependent behavior; and changes under the signal identify recommendation compliance. These observations can be compared with the computational benchmark and can reveal whether the comparison policy or signal interface needs revision. Classroom choices are treated as exploratory behavioral evidence rather than a population-level effect.

5 Advanced development, real-world impact, and roadmap

The project integrates strategic prediction, dynamic computation, and behavioral testing within one transaction-priority problem. Academically, it distinguishes individual best responses from allocation and welfare outcomes and identifies a trade-off between better priority coordination and higher fee expenditure. For industry, the comparison may inform wallet or transaction-priority interfaces; for users, it makes visible the cost of obtaining better priority alignment. If externally validated, nonbinding recommendations may provide a simple coordination tool when users differ in urgency, but improved allocation should not be interpreted as an automatic welfare gain.

The main risks are stylized payoffs, unequal effects across user types, and overextending a two-user model to a live blockchain market. We therefore report type-specific outcomes and treat Algorand as the motivating environment rather than empirical evidence. If behavioral evidence contradicts the benchmark, we will revise the comparison policy or signal design and rerun the computational tests. The next study will test participant beliefs and signal compliance under alternative fee menus and payoff calibrations.

Expected 2056 Nobel Prize and Turing Award contribution statement. We aspire to develop transparent and testable mechanisms for allocating scarce digital resources under private information.

Open Science Statement

The GitHub repository records the project files and implementation at final verified commit `8487e72`, while the Colab reproduces the strategic benchmark, three-condition simulation, round-level results, and signal-compliance sensitivity analysis. The Hugging Face Space implements the corresponding behavioral game. Seeds, parameter definitions, and artifact versions are recorded so computational results can be reproduced.

Statement of Contribution to the UN's SDGs

The project relates to SDG 9 through the design of reliable digital infrastructure [7]. Its potential contribution is narrower than a demonstrated SDG impact: if validated, better priority-allocation interfaces may reduce coordination failures while preserving access for users with different urgency levels.

Author Notes

Acknowledgements

We thank Professor Luyao Zhang for her guidance on interdisciplinary research design and the integration of game theory, social choice, and mechanism design; Professor Kenneth S. (Ken) Rogerson for his instruction in economic reasoning; and Han Zhang for helping us clarify our research gap and distinguish our contribution from existing studies.

We are grateful to the peer reviewers and discussion partners whose feedback informed our revisions following the September 28, 2026 symposium. Gihun Lee (FP5) provided formal handwritten peer evaluation, emphasizing the need to distinguish nonbinding coordination signals from changes in the information environment. Longfei Jing (FP6) questioned whether simulated user behavior could represent real decisions and recommended human-participant validation through the Hugging Face game. Mustafa Ayub Khan (FP5) raised concerns about the fee-choice design and the lack of human behavioral evidence, recommending further testing of the behavioral game. On October 12, 2026, Muhan Chen also provided a literature-based review of three closely related studies, helping us distinguish our nonbinding coordination approach from existing research on fee competition and payment-mechanism redesign.

We also acknowledge Tencent's Shanghai office and the Shanghai Science and Technology Museum for providing opportunities to explore technological innovation and its real-world applications, as well as the COMSCI/ECON 206 classroom community for constructive discussions, peer feedback, and interdisciplinary learning.

The PS2 annotated template, Week 4 strategic-game materials, and Week 5 auction and allocation materials helped us check the game structure, solution concept, and allocation interpretation. The computational analysis was developed using Python and Google Colab, with GitHub maintaining the project repository and outputs. Hugging Face Spaces hosts the behavioral game, and the A0 poster was developed from the course template. OpenAI Codex and ChatGPT (GPT-5.6 Sol) were used as disclosed in Appendix A.1. These tools did not replace the authors' decisions about the research question, model, assumptions, or interpretation. The authors remain responsible for the final work and any remaining errors.

Team and individual contributions

Lin Zhang and Zijie Wang jointly developed the central research question of how private transaction urgency affects the allocation of scarce transaction priority. In Version 1, the team jointly decided to represent this question as a one-shot High/Low fee-choice game and to compare complete information, incomplete information, and a nonbinding coordination signal. Zijie Wang developed the initial proposal framework, computational implementation, and GitHub repository. Lin Zhang revised and integrated the proposal, developed the Hugging Face behavioral game, and prepared the A0 poster. Both authors reviewed the model and its interpretation across the linked artifacts.

After the September 28 symposium, the authors jointly reviewed the peer comments and decided that Version 2 should restore dynamic waiting while preserving discrete fee choice and private urgency. The revised model therefore uses three connected rounds and Low, Medium, and High fees, with waiting carried across rounds for unconfirmed users. The authors also separated the stage Bayesian equilibrium prediction from the state-responsive policy used for computational comparison.

Zijie Wang first revised the Hugging Face game to implement the three-round decision environment, private Urgent/Patient types, $L/M/H$ fee choices, waiting transitions, and the three information conditions. Lin Zhang then revised the proposal and Colab so that the formal model, strategic benchmark, simulation policy, signal rule, metrics, and evidence boundary matched the behavioral game. She reran and checked the computational comparison, including the added fee-expenditure measure and signal-compliance sensitivity analysis. Zijie Wang reviewed the revised model and computational outputs and then updated the GitHub materials to reflect the agreed Version 2 design. They subsequently revised the poster so that the presentation uses the same research question, model, results, and limitations.

Both authors are responsible for the final integrated argument. Their remaining responsibility is to verify that the final proposal, GitHub repository, Colab, Hugging Face Space, and poster preserve the same Version 2 assumptions, outputs, and evidence status.

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A Technical details and reproducibility

Version 2 implements a three-round dynamic fee-choice game with two users. Each user has a fixed private type $\theta_i \in \{U, P\}$ and in each round $t \in \{1, 2, 3\}$ chooses

$$a_{i,t} \in \{L, M, H\}.$$

The baseline prior is $\Pr(U) = 0.6$. Fee costs are $c(L) = 1$, $c(M) = 2$, and $c(H) = 3$, with $(V_U, d_U) = (8, 2)$ and $(V_P, d_P) = (5, 1)$. The higher fee receives confirmation priority and equal fees use a 50–50 random tie-break. Exactly one user is confirmed in each round. A confirmed user's waiting state resets to zero, while an unconfirmed user's waiting state increases by one.

The strategic benchmark is separate from the simulation policy. Against an opponent choosing H, a lower fee gives

$$-d_\theta(w + 1),$$

whereas choosing H gives

$$\frac{1}{2}[V_\theta - c(H)] - \frac{1}{2}d_\theta(w+1).$$

For both types and all waiting states reached in the three-round model, H is a strict best response to H. The Colab verifies this condition directly and therefore reports (H, H) as a stage Bayesian Nash equilibrium rather than claiming to solve a dynamic equilibrium for the full three-round game [5].

For the computational comparison, the state-responsive policy is

$$\pi(U,w) = \begin{cases} M, & w = 0, \\ H, & w \geq 1, \end{cases} \qquad \pi(P,w) = \begin{cases} L, & w = 0, \\ M, & w = 1, \\ H, & w \geq 2. \end{cases}$$

The complete-information benchmark coordinates differentiated recommendations using both users' types and waiting states. Under incomplete information, each simulated user follows the comparison policy using only its own state. The signal condition uses

$$q_{i,t} = 21\{\theta_i = U\} + w_{i,t}$$

to generate differentiated, nonbinding recommendations using both users' states without revealing either user's private type to the other user. Baseline signal compliance is $\rho = 0.85$. A small tremble probability is retained in the simulation so that recommendations do not mechanically determine every action.

The simulation records same-fee collision, priority alignment, urgent-user confirmation, waiting, fee expenditure, individual utility, and total user welfare. Fee expenditure is the fee paid by the confirmed user; because exactly one transaction is confirmed in each round, mean fee expenditure per round is also mean fee expenditure per confirmed transaction. In 100,000 simulated sessions, incomplete information produces a same-fee collision rate of 33.36%, priority alignment of 87.11%, urgent-user confirmation of 66.39%, mean fee expenditure of 2.352, and mean total user welfare per round of 3.057. With the signal, the corresponding values are 5.20%, 97.99%, 81.25%, 2.812, and 2.678. The result therefore identifies an allocation–payment–welfare trade-off rather than an unconditional welfare improvement.

The signal-compliance sensitivity check varies ρ from 0 to 1. As compliance rises, same-fee collision falls and priority alignment rises, while mean fee expenditure increases and mean user welfare declines under the baseline calibration. These outputs are model-generated simulation results. They are not participant data and are not evidence of current Algorand network performance.

The runnable Colab contains the parameter definitions, strategic-benchmark verification, session simulation, condition-level and round-level summaries, plots, and signal-compliance sensitivity analysis. The computational files and outputs are maintained in the GitHub repository at commit `8487e72`. The Hugging Face Space implements the corresponding three-round behavioral task, and the final A0 poster reports the same model and computational evidence. A fresh run of the final notebook is used to verify all numerical claims before submission.

A.1 AI-use disclosure

The central PS2 research idea and the decisions separating Version 1 and Version 2 were developed through discussion between the two authors.

OpenAI Codex was used during development to assist with coding and document implementation. Its main uses included transferring author-written material into the required LaTeX structure, debugging and organizing parts of the computational implementation, and refining visual materials.

ChatGPT using GPT-5.6 Sol was used during later revision to improve wording, organize author-written material into LaTeX, check consistency among the proposal and linked artifacts, and assist with selected computational and visual revisions. In these interactions, the authors supplied the research design, formulas, computational results, and intended interpretation, then reviewed and edited the resulting material.

AI-generated suggestions were treated as drafting and implementation assistance. The authors checked the final strategic model, payoff definitions, equilibrium interpretation, computational outputs, citations, and evidence-status statements against the proposal and linked artifacts. Lin Zhang and Zijie Wang remain responsible for all final research decisions, claims, computations, and submitted materials.

B Cumulative Intellectual Development

The project first developed from a transaction-timing idea in which users chose whether to Submit or Wait under congestion. The early proposal treated transaction purpose as private information and linked individual timing decisions to aggregate load and later congestion states. Work on this model introduced the broader question that has remained throughout PS2: how users with different transaction urgency interact when transaction-processing priority is scarce. Related work on blockchain transaction timing, capacity, and strategic decision-making informed this early stage [2, 4, 8].

During the first PS2 revision, the team simplified the strategic representation to isolate private information and coordination. Version 1 therefore used a one-shot High/Low fee-choice game. Two users privately observed whether they were Urgent or Patient and chose simultaneously. Same-fee choices were treated as coordination failure, while differentiated choices assigned different priority levels. The Colab compared complete information, incomplete information, and incomplete information with a nonbinding coordination signal. The Hugging Face game, proposal, and poster were revised around the same binary action set. This version made the incomplete-information benchmark easier to state and made the role of private urgency more visible.

The September 28 peer review identified a new problem with this simplification. The one-shot model no longer represented the waiting dynamics that motivated the transaction setting, and the relationship among the strategic prediction, computational policy, signal condition, and behavioral game needed to be clearer. The team therefore decided to retain discrete fee choice and private urgency but restore the dynamic state.

Version 2 uses the same two users across three connected rounds. Each user chooses L, M, or H, and an unconfirmed user accumulates waiting into the next round. Equal fees no longer represent literal

congestion; instead, they create a priority tie resolved randomly. This distinction separates the strategic coordination problem from its dynamic consequence.

A second Version 2 revision separated two objects that Version 1 had treated too closely. The game-theoretic prediction now uses a stage Bayesian best-response benchmark: under the baseline calibration, H is a strict best response to H for both types, so (H, H) is a stage Bayesian Nash equilibrium. The dynamic simulation instead uses a transparent type-and-wait comparison policy. The policy is therefore a computational benchmark, not an equilibrium claim.

The revised computation also changed the interpretation of the mechanism. The signal substantially reduces same-fee collision and improves priority alignment and urgent-user confirmation, but it increases mean fee expenditure and lowers total user welfare under the baseline calibration. The intellectual consequence is that the project no longer asks only whether coordination can be improved. It now makes explicit the trade-off between priority allocation, payment, and user welfare.

The Version 2 Hugging Face game, Colab, proposal, teaser, and poster were then aligned around this same dynamic model. Behavioral evidence remains exploratory: the interface records beliefs, confidence, reasoning, and choices, while the simulation results remain model-generated. This preserves the distinction between a strategic prediction, a computational comparison, and evidence about actual behavior.

C Open-source and three-lens inspiration

Algorand remains the motivating transaction environment rather than an empirical dataset. Its protocol literature provides the broader blockchain setting [3], while transaction-fee mechanism design motivates the allocation of scarce processing capacity through user payments [6]. The present model deliberately replaces the live network with two users, three discrete fee levels, and three rounds. Its numerical results therefore describe the model rather than current Algorand performance.

The open computational workflow makes this abstraction inspectable. Google Colab provides the runnable environment for the strategic-benchmark check, simulation, condition comparison, and sensitivity analysis. GitHub records the computational materials and final outputs at a fixed commit. Hugging Face Spaces provides the behavioral interface in which a player experiences the same private type, fee choices, waiting transition, and information conditions. The A0 poster communicates the same research question and evidence boundary. These artifacts serve different purposes but use the same Version 2 model.

The three disciplinary lenses impose different tests on that model. Game theory asks how users respond strategically under private information and supplies the stage Bayesian best-response benchmark [5]. Social choice treats confirmation priority as the scarce resource and evaluates whether allocation reflects urgency while making the trade-off between fee expenditure and user welfare visible. Mechanism design changes the information environment through a nonbinding coordination signal and tests whether differentiated recommendations improve allocation without revealing the other user's private type or forcing an action.

The computational comparison shows why all three lenses are needed. The signal improves priority alignment but also increases fee expenditure and lowers user welfare under the baseline calibration. An equilibrium prediction alone would not reveal this allocation trade-off, and an allocation comparison alone would not establish how actual users respond to the recommendation. The behavioral artifact therefore provides the planned link between strategic beliefs and mechanism performance.

D Review response and revision record

During the September 28 symposium and peer-review session, we received several comments that informed the subsequent revision.

One suggestion was to make the relationship among the proposal, computational analysis, and behavioral game more consistent. In Version 1, the one-shot High/Low model simplified the strategic problem but no longer represented the waiting dynamics that motivated the transaction setting. After discussing this feedback, we revised the project around one common three-round fee-choice environment. The proposal, Colab, and Hugging Face game now use the same Urgent/Patient types, $L/M/H$ actions, waiting-state transitions, and information conditions.

Another comment concerned the distinction between the complete-information benchmark and the coordination signal. We clarified that the complete-information benchmark can coordinate actions using both users' type and waiting information. Under the signal condition, the other user's private type is not revealed; instead, the system provides a differentiated but nonbinding recommendation. This distinction is now stated consistently in the proposal, Colab, and Hugging Face game.

Peer feedback also emphasized the need to connect the project more clearly to game theory, social choice, and mechanism design. In response, the revised proposal separates their roles. Game theory provides the stage Bayesian best-response benchmark, social choice evaluates how scarce transaction priority is allocated and the resulting welfare trade-off, and mechanism design studies whether a nonbinding signal can improve coordination. The computational analysis was also expanded to report fee expenditure as a payment measure, making the trade-off between priority alignment, payment, and user welfare explicit.

A further suggestion was to strengthen the project with actual behavioral or real-world data rather than relying only on simulation. We agree that this would provide stronger evidence, but collecting and validating sufficient participant or live-network data is difficult within the current project scope. We therefore do not present the simulation or Hugging Face game as population-level behavioral or empirical evidence. A future extension could collect participant choices or suitable transaction data and compare them with the strategic and computational benchmarks developed here.

Following these comments, we reran the revised Colab and checked the numerical results against the claims reported in the proposal. We also updated the linked artifacts so that the current Version 2 model and its evidence limitations are stated consistently, while retaining Version 1 in the appendices as a record of the project's development.

E Structured abstract and cumulative proposal map

The structured abstract preserves both completed PS2 versions. Table 1 records the September 27 Version 1 design as submitted for review. Table 2 records the Version 2 model after the September 28 review and subsequent computational verification. Version 1 is retained as an intellectual-development record rather than presented as the current model.

Table 1: Version 1 structured abstract and cumulative proposal map.

Proposal element	Current statement / change
Background	Transaction priority is scarce, and users choosing fees without observing others' urgency face a coordination problem. Existing work provides foundations for Nash and Bayesian equilibrium, coordination, and transaction-fee design. The present project isolates a narrower question: whether two privately informed users can differentiate High and Low fee choices when matching choices create congestion.
Motivation and application	The motivating case is a simplified transaction-fee interface in which an Urgent and a Patient user may value priority differently. Failed coordination can place both users at the same fee level and produce congestion. The simplified setting allows the role of private information and coordination signals to be studied before introducing more users, continuous fees, or live-network dynamics.
Three connected questions	Economics asks how private urgency changes equilibrium, priority allocation, and welfare. Computation asks how the Bayesian benchmark and information treatments can be solved and compared reproducibly. Behavioral science asks whether actual users follow the type-dependent benchmark and whether a nonbinding signal changes their decisions. Together these questions ask when decentralized fee choices can successfully allocate scarce transaction priority under private information.
Method and model assumptions	Two users privately observe type $\theta_i \in \{U, P\}$ and simultaneously choose $a_i \in \{H, L\}$. Matching actions receive the congestion payoff, while differentiated actions assign High and Low rewards with $R_H > R_L > R_C$. The baseline calibration gives Urgent and Patient users different rewards and produces the pure Bayesian strategy $U \to H,\ P \to L$. The Colab compares a coordinated complete-information benchmark, incomplete information, and incomplete information with a nonbinding signal.
Results or expected results	The completed computational analysis verifies the baseline Bayesian equilibrium and shows a large coordination loss under incomplete information. In the baseline simulation, same-fee congestion is about 51.5% under incomplete information and about 12.6% with the coordination signal; mean total welfare increases from about 6.78 to 9.76. These are model-generated results. The planned behavioral test asks whether human choices and signal responses follow the same pattern.
Intellectual merits	The project separates three questions that are easy to combine incorrectly: what privately informed users should do under the model, whether decentralized choices produce an efficient allocation, and whether people actually follow the Bayesian benchmark. Linking equilibrium analysis, reproducible simulation, and a behavioral interface makes these distinctions directly testable within the same strategic game.
Practical impacts	If the mechanism is externally validated, the results may inform how transaction interfaces present discrete fee choices or nonbinding coordination recommendations. Possible benefits include lower congestion and better use of scarce priority, while risks include unequal access for Patient or lower-value users and oversimplification of a live network. The project therefore relates to SDG 9 through reliable digital infrastructure, but no practical impact is claimed from the current simulation alone.
Feedback received and revision made	The project initially used a repeated Submit/Wait congestion game. Internal review of the strategic model and assignment requirements led us to retain the transaction-priority question but replace the dynamic structure with a one-shot High/Low Bayesian anti-coordination game. The Colab, Hugging Face interface, teaser, and poster were then revised around the same types, actions, information conditions, and evidence boundary. Formal symposium feedback will be added after September 28.

Table 2: Version 2 structured abstract and cumulative proposal map.

Proposal element	Current statement / change
Background	Transaction priority is scarce, and users with different urgency may choose fees without observing one another's private state. Existing work provides foundations for Bayesian equilibrium and transaction-fee mechanism design. The revised project studies a dynamic setting in which two privately informed users choose among Low, Medium, and High fees across three connected rounds, while unconfirmed users accumulate waiting.
Motivation and application	The motivating case remains a simplified transaction-fee interface in which Urgent and Patient users value confirmation differently. Higher fees receive priority, while equal fees require a random tie-break and unconfirmed users accumulate delay. The revised setting allows us to study how private urgency, waiting, fee expenditure, and coordination jointly affect transaction-priority allocation.
Three connected questions	Economics asks how urgency, waiting, and fee costs shape strategic choices and welfare. Computation asks how information and coordination rules change allocation outcomes. Behavioral science asks how users' beliefs about the opponent affect fee choices and responses to recommendations. Together these questions ask how private urgency and accumulated waiting affect transaction-priority allocation and whether a coordination signal can improve it.
Method and model assumptions	Two users have fixed private types $\theta_i \in \{U, P\}$ and choose $a_{i,t} \in \{L, M, H\}$ in each of three rounds. Higher fees receive confirmation priority, equal fees use a 50–50 tie-break, and unconfirmed users accumulate waiting. The strategic prediction uses a stage Bayesian best-response benchmark, while the Colab separately applies a state-responsive comparison policy under complete information, incomplete information, and incomplete information with a nonbinding signal.
Results or expected results	In 100,000 simulated sessions, the signal reduces same-fee collision from 33.36% to 5.20% and raises priority alignment from 87.11% to 97.99%. Urgent-user confirmation rises from 66.39% to 81.25%, while mean fee expenditure rises from 2.352 to 2.812 and mean total user welfare per round falls from 3.057 to 2.678. These are model-generated results; behavioral evidence remains exploratory.
Intellectual merits	The revised project separates the stage strategic prediction from the dynamic computational comparison and evaluates allocation, payment, and welfare within the same model. Linking game theory, social choice, mechanism design, reproducible simulation, and a behavioral interface shows that better priority coordination can coexist with higher fee expenditure and lower user welfare.
Practical impacts	If externally validated, the results may inform how transaction interfaces present discrete fee choices or nonbinding coordination recommendations. Possible benefits include better alignment of scarce priority with user urgency, while risks include higher fee expenditure, unequal effects across user types, and oversimplification of a live network. The project therefore does not infer practical effectiveness from simulation alone.
Feedback received and revision made	Following the September 28 peer review, we revised the one-shot High/Low model into a three-round Low/Medium/High fee-choice game with accumulated waiting. We clarified the distinction between complete information and the nonbinding signal, strengthened the three-lens integration, added fee expenditure to the computational evaluation, and aligned the proposal, Colab, Hugging Face game, teaser, and poster around the same Version 2 model.

F PS2 extension record: allocation, linked artifacts, poster, and symposium

F.1 Allocation record

The Version 2 application is an allocation problem. The scarce resource is confirmation priority. Two users with private Urgent or Patient types choose among L, M, and H in each of three rounds. The higher fee receives priority and an equal-fee tie is resolved randomly. The confirmed user pays the selected fee; there is no separate seller bidding stage or auction stopping rule. The next round begins after confirmation and the waiting state is updated.

The rational strategic benchmark and the allocation comparison answer different questions. The stage best-response calculation shows that H is a strict best response to an opponent choosing H under the baseline calibration. The computational comparison then holds the priority rule fixed while changing the information and coordination environment. Complete information coordinates differentiated recommendations using both users' states. Under incomplete information, each user follows the comparison policy using only its own state. The signal condition retains private types between users but adds a differentiated, nonbinding recommendation.

The comparison reports allocation, payment, utility, and welfare outcomes. Relative to incomplete information, the signal reduces same-fee collision from 33.36% to 5.20%, raises priority alignment from 87.11% to 97.99%, and raises urgent-user confirmation from 66.39% to 81.25%. Mean fee expenditure increases from 2.352 to 2.812, while mean total user welfare per round decreases from 3.057 to 2.678. The mechanism-design implication is therefore conditional: coordination improves allocation but creates a payment and welfare cost under the stated calibration.

The signal-compliance sensitivity analysis provides a robustness check. Increasing compliance moves outcomes toward the coordinated benchmark: collision falls and priority alignment rises, but fee expenditure also rises and user welfare declines. This comparison supports retaining the signal as a coordination treatment while rejecting the stronger claim that it necessarily improves welfare.

F.2 Linked-artifact parity

Table 3: Version 2 linked-artifact parity check.

Artifact	Current status and connection
GitHub	Contains the Version 2 computational materials, outputs, and documentation corresponding to the final proposal. Final verified commit: `8487e72`.
Google Colab	Implements the stage Bayesian best-response check, three-round simulation, complete/incomplete/signal comparison, round-level outputs, fee-expenditure measure, and signal-compliance sensitivity analysis. Its results are model-generated rather than participant evidence.
Hugging Face	Implements the same three-round private-type $L/M/H$ decision environment. It records beliefs, confidence, reasoning, and fee choices and includes the complete-information, incomplete-information, and nonbinding-signal conditions. Classroom use provides exploratory behavioral evidence only.
A0 poster	Uses the same shared question, Version 2 strategic model, three disciplinary lenses, computational results, behavioral design, allocation–payment–welfare tradeoff, practical implications, and evidence limitations as the proposal.

F.3 Symposium and practical-impact record

The September 28 symposium provided the review step connecting Version 1 to Version 2. The team retained the transaction-priority application and private Urgent/Patient types, but revised the strategic representation, computational comparison, and artifact descriptions in response to peer review. The final revision preserves the original Version 1 record in Appendix E while using Version 2 consistently in the main proposal and current artifacts.

The practical case is a transaction-priority interface that may provide nonbinding recommendations when users differ in urgency. Users are directly affected through confirmation delay and fee expenditure; interface and protocol designers determine how priority choices and recommendations are presented. The potential benefit is better alignment of scarce priority with urgency. The main risks are higher payments, unequal effects across user types, and treating results from a stylized two-user simulation as evidence about a live network.

The current evidence does not justify deployment or a population-level behavioral claim. Before a practical recommendation, the payoff calibration would require external validation and participant behavior would need to be compared with the strategic and computational benchmarks. If behavioral evidence contradicts the assumed comparison policy or signal response, the correction route is to revise the policy or interface, rerun the computational comparison, and update the linked artifacts consistently.